

EXP 1.1

Program:

```
clc;
clear;
close all;
Fs = 10000;
Fp = 1000;
Fst = 1500;
Rp = 1;
Rs = 60;
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);
[N,Wn] = buttord(Wp,Ws,Rp,Rs)
[b,a] = butter(N,Wn,'low')
figure;
freqz(b,a);
figure;
impz(b,a);
figure;
x = ones(1,50);
y = filter(b,a,x);
stem(y);
figure;
zplane(b,a);
```

Output:

```
N =  
    17  
  
Wn =  
    0.2083  
  
b =  
    1.0e-05 *  
Columns 1 through 15  
    0.0000    0.0005    0.0038    0.0189    0.0662    0.1721    0.3442    0.5409    0.6761    0.6761    0.5409    0.3442    0.1721    0.0662    0.0189  
Columns 16 through 18  
    0.0038    0.0005    0.0000  
  
a =  
Columns 1 through 15  
    1.0000   -9.9105   47.1991  -143.1269  308.9374  -502.9610  638.7325  -645.6106  525.3425  -345.8305  184.0468  -78.6851   26.6787   -7.0197    1.3836  
Columns 16 through 18  
   -0.1924    0.0169   -0.0007
```



